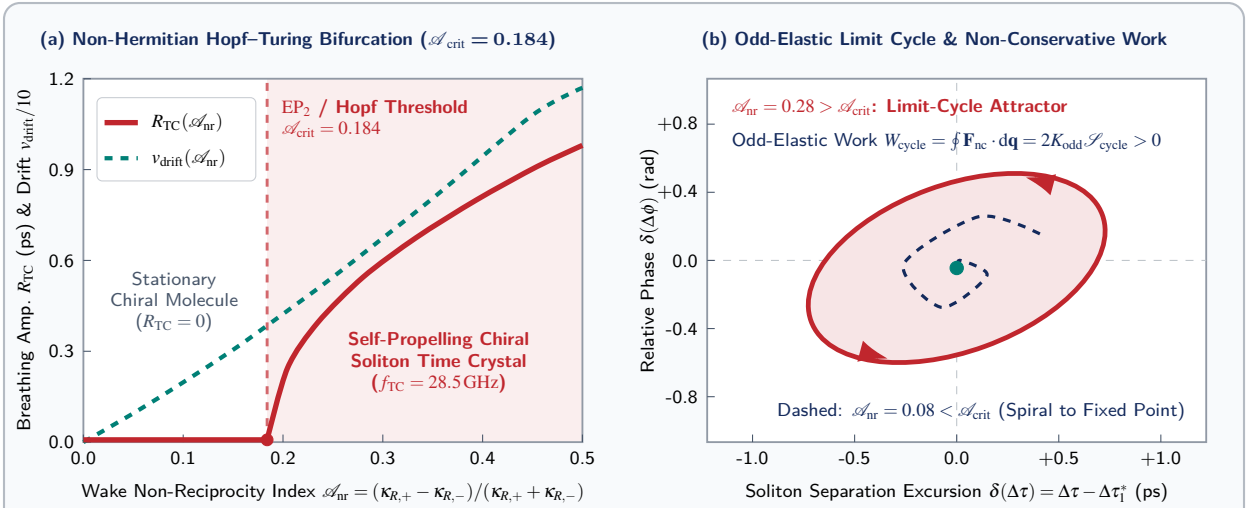
(a) Non-Hermitian Hopf–Turing Bifurcation ($\mathscr{A}_{\rm crit} = 0.184$)
$R_{\rm TC}(\mathscr{A}_{\rm nr})$
$v_{\rm drift}(\mathscr{A}_{\rm nr})$
EP$_2$ / Hopf Threshold
$\mathscr{A}_{\rm crit} = 0.184$
Stationary
Chiral Molecule
($R_{\rm TC} = 0$)
Self-Propelling Chiral
Soliton Time Crystal
($f_{\rm TC} = 28.5\,$GHz)
Breathing Amp. $R_{\rm TC}$ (ps) & Drift $v_{\rm drift}/10$
Wake Non-Reciprocity Index $\mathscr{A}_{\rm nr} = (\kappa_{R,+} - \kappa_{R,-})/(\kappa_{R,+} + \kappa_{R,-})$
0.0
0.3
0.6
0.9
1.2
0.1
0.2
0.4
0.5
(b) Odd-Elastic Limit Cycle & Non-Conservative Work
$\mathscr{A}_{\rm nr} = 0.28 > \mathscr{A}_{\rm crit}$: Limit-Cycle Attractor
Odd-Elastic Work $W_{\rm cycle} = \oint \mathbf{F}_{\rm nc} \cdot d\mathbf{q} = 2K_{\rm odd}\mathscr{S}_{\rm cycle} > 0$
Dashed: $\mathscr{A}_{\rm nr} = 0.08 < \mathscr{A}_{\rm crit}$ (Spiral to Fixed Point)
Relative Phase $\delta(\Delta\phi)$ (rad)
Soliton Separation Excursion $\delta(\Delta\tau) = \Delta\tau - \Delta\tau_1^*$ (ps)
+0.8
+0.4
0.0
-0.4
-0.8
-1.0
-0.5
0.0
+0.5
+1.0